

Jason White | Robotacist

☎ 510-359-8681 @jrw5690@gmail.com 🔥 jwhite5497 🐼 jw18dn

Employment

Foundation Industries

2026 - present

Senior Controls and Software Engineer (C++ & Python)

- o Led controls and software development for a 22-DOF tendon driven hand, scaling it from a single motor prototype to a modular control stack deployed on humanoid hardware [hardware, ~500Hz] <https://www.youtube.com/watch?v=Nhr7ZnFZYkA&t=1s>
- o Increased hardware communication rate from 20 Hz over UART to 250 Hz over two CANFD lines through hardware abstraction, controller optimization, & firmware development
- o Developed a highly accurate open loop position and torque dependent state estimator with a 10 microsecond solve time
- o Improved the Phantom whole body inverse kinematics (optimization fixes), Inspire hand grasp behavior (admittance control), and WBC using NVIDIA's Sonic
- o Streamlined calibration routines, zeroing routines, and added safety controllers
- o Design decisions on tactile & position sensing, PCB components/locations, and communication protocol

Ghost Robotics

2024 - 2026

Controls and Software Engineer (C++ & Python)

- o Developed a system-wide locomotion controller using Model Predictive Control (MPC) and Pinocchio for real-time dynamic control [hardware, ~500Hz]
- o Improved blind locomotion robustness on rugged terrain, reducing falls/flips from **7/20 to 1/20 trials** without reliance on perception [hardware, ~500Hz]
- o Developed point cloud clustering and 2D height map filtering to enable terrain-aware locomotion [hardware, ~90Hz]
- o Refactored core components of the codebase, eliminating blind locomotion low-level crashes and improving system reliability [hardware, ~500Hz]
- o Designed filtering and tracking pipelines for manipulator input signals, eliminating drift and improving control stability [hardware, ~500Hz]
- o Led system identification and design of next-generation leg actuators and drivetrain, achieving **3x torque output** and **50% mass reduction** [hardware]

Optimal Robotics Lab

2018 - 2023

Graduate Research Assistant (Matlab & Python)

- o Obstacle avoidance algorithm: for quadrotor & bipedal avoidance of fast-moving obstacles [hardware, ~1kHz] <https://www.youtube.com/watch?v=cKKLuRdrqcQ>
- o Bipedal balancing controller: for dynamic terrain tested on Cassie [hardware, >2kHz] <https://www.youtube.com/watch?v=q10Wg61H4UQ>
- o Real-time reactive planning and control for monopods (with flexible contact sequencing) [simulated, ~500Hz] <https://www.youtube.com/watch?v=EquyU9cm-SM>
- o Quadrupedal stepping controller: using optimal control techniques (preliminary results) [simulated, ~500Hz] www.linkedin.com/feed/update/urn:li:activity:7122585426162249730/

Technical Skills

Languages: C++, Python, MATLAB

Simulations: MuJoCo, Simulink, Bullet

Control Methods: Operational Space Control, Whole-Body Control, Forward/Inverse Dynamics, Joint Space Control, PD/PID Control, Force Control

Planning: Model Predictive Control, Direct Collocation, Sampling-based methods (RRT*, A*)

Optimization: Linear/Quadratic/Nonlinear Programming, Newton-Raphson

Awards, Fellowships, and Honors

2022: NASA Big Idea Challenge Finalist (ET-QUAD)

NASA

2022: Best Locomotion Paper, IEEE International Conference on Robotics and Automation

IEEE

2021: Research Fellowship Recipient, DEVCOM Army Research Laboratory

National

Peer Reviewed Publications

Jason White, David Jay, Tianze Wang, and Christian Hubicki, "Avoiding Dynamic Obstacles with Real-time Motion Planning using Quadratic Programming for Varied Locomotion Modes," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 13626-13633) IEEE.

Jason White, Max Austin, Jonathan Clark, Christian Hubicki, and Jason Pusey. "Online Gait Optimization for Running in Resistive Media," in *2022 7th International Conference on Robotics and Automation Engineering (ICRAE)* (pp. 282-286). IEEE.

Jason White, Dylan Swart, and Christian Hubicki, "Force-based Control of Bipedal Balancing on Dynamic Terrain with the "Tallahassee Cassie" Robotic Platform," in *2020 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 6618-6624). IEEE.

Tianze Wang, **Jason White**, and Christian Hubicki, "Real-time Dynamic Bipedal Avoidance," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* IEEE.

Adwait Mane, Dylan Swart, **Jason White**, and Christian Hubicki. "Trajectory Optimization Formulation with Smooth Analytical Derivatives for Track-leg and Wheel-leg Ground Robots." in *2022 International Conference on Robotics and Automation (ICRA)* (pp. 5762-5768). IEEE.

Max Austin, John Nicholson, **Jason White**, Sean Gart, Ashley Chase, Jason Pusey, Christian Hubicki, Jonathan Clark, "Optimizing Dynamic Legged Locomotion in Mixed, Resistive Media," in *2022 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)* (pp. 1482-1488). IEEE.

Manuscripts & Posters

Jason White and Christian Hubicki, "Through-contact Monopod Motion Generation via MPC: Convex Formulation and Locomotion Analysis." 2024.

Jason White and Christian Hubicki, "Real-time Adversarial Obstacle Avoidance." 2023.

Jason White and Christian Hubicki "Toward Reduced-Order Modeling Approaches for Rapidly Reactive Locomotion." 2021 Dynamic Walking

Education

Florida State University

Ph.D. Mechanical Engineering - Legged Robot Control

M.S. Mechanical Engineering - Robotics, Dynamics, & Controls

Advisor: Christian Hubicki Ph.D.

Virginia Polytechnic Institute and State University

B.S. Civil Engineering - Structural Focus